GITAM Deemed to be University

Biotechnology | Course Code: BT301

Genetic Engineering & Recombinant DNA - Recombinant DNA Technology & Geneti

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Overview: Type II restriction enzymes, pUC19 and pBR322 vectors, Sanger sequencing, Taq DNA pol

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Molecular Tools & Enzymes
Restriction endonucleases, DNA ligases, Alkaline phosphatases, Reverse transcriptase, Polynucleotide

Unit 2: Cloning Vectors & Transformation
Plasmids pBR322, pUC19, Bacteriophage lambda, Cosmids, YAC, BAC, Competent cell preparation, Electro

Unit 3: Gene Amplification & Sequencing
Polymerase Chain Reaction PCR variations RT-PCR, qPCR, Sanger chain termination sequencing, Next-Gen

Unit 4: Genome Editing & Applications
CRISPR-Cas9 target recognition, sgRNA design, Zinc Finger Nucleases, Recombinant therapeutics Insuli

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Describe the key structural components of pBR322 cloning vector and explain insertional inactivation f
2. Outline the thermal cycling phases of Polymerase Chain Reaction PCR and calculate theoretical amplicon
3. Explain the mechanism of CRISPR-Cas9 mediated double-strand break repair via NHEJ and HDR pathways.

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* T.A. Brown - Gene Cloning and DNA Analysis: An Introduction Wiley-Blackwell 7th Ed
* Bernard R. Glick, Cheryl L. Patten - Molecular Biotechnology: Principles and Applications of Recom

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